Does Remote Sensing Beat the Trail Map? Predicting Hiking Speed in Unseen Regions from Lidar Terrain and OpenStreetMap Tags

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***Abstract*—[TODO: Draft by Claude: rewrite in your own words.] Estimates of walking time on trails still rely on Tobler's 1993 hiking function. We ask what corrects it in regions a model has never seen, using 290,640 walking traversals of 50 m trail segments from public OpenStreetMap (OSM) GPS traces in 17 North American mountain regions. Holding out one region at a time and scoring calibrated predictive distributions (CRPS), we find Tobler's function 12.5% worse than a refitted slope curve. Terrain from elevation models, mostly airborne lidar, improves CRPS by 2.1–2.8% in unseen regions and beats OSM trail tags in 16 of 17; tags add 0.4–0.9%, and lidar vegetation and satellite embeddings help mainly where the model was trained. For the same hiker on the same trail, descending is 8–23% faster than climbing, not 42% as Tobler implies. A one-input terrain-adjusted Tobler function keeps part of the gain, but a hiker's own pace still matters most over whole hikes.**

***Index Terms*—hiking speed, lidar, OpenStreetMap, GPS trajectories, transfer, calibration**

I. Introduction

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How long a walk will take matters for search and rescue, for wildland firefighters' escape routes [1] and for the times trail apps show. A common starting point is Tobler's hiking function [2], $v = 6\exp(-3.5\,|s + 0.05|)$ km/h on slope s, which peaks at 6 km/h on a gentle descent. It is still in use, from maps of what a person can reach on foot [3] to traverse planning for planetary exploration [4].

Crowdsourced GPS tracks show that the function is too fast and too peaked, and that speeds at one slope vary widely [5], [6]. Newer models add covariates: OSM terrain classes and lidar-measured obstruction in England and Wales [7]; boosted trees on 16 factors, weather included, in Hong Kong [8]; a latent walking speed predicted from Sentinel-2 imagery in one Croatian range [9]; the hiker's pace on the first part of a hike [10]. STRIDE [11] models walking speed from lidar slope, understory density [12] and ground roughness, calibrated on timed field walks. On roads, speeds and their spread have been learned from overhead imagery and lidar [13], [14].

Wood et al. [7] test with 10-fold cross-validation and Huang et al. [8] with a random split, so their test tracks come from the areas they trained on. Yet predictions matter most where nobody has recorded a hike: in the White Mountains, one of our best-sampled regions, 77% of the 3,369 km of mapped trail has no trace in our data. Two sources could describe such trails. On the trail map, OSM mappers tag difficulty (sac_scale), surface and visibility, but unevenly: sac_scale is on 1–96% of walked trail, depending on the region. Remote sensing reaches every trail, tagged or not: the USGS 3D Elevation Program collects airborne lidar across the United States [15], and satellite embeddings summarize imagery worldwide [16]. To our knowledge, no study has tested which of the two works in regions a model has never seen.

We test it on public OpenStreetMap GPS traces [17] from 17 mountain regions of the United States and Canada. We contribute (i) a dataset of 290,640 walking traversals of 50 m trail segments (1,992 outings) with slope, OSM tags and remote-sensing covariates (Sec. II); (ii) a transfer test that holds out whole regions, zones or half the continent and scores predictive distributions with the continuous ranked probability score (CRPS) [18], against published hiking functions (Sec. III); and (iii) the finding that terrain measured from elevation data transfers ($+2.1$ to $+2.8\%$ CRPS skill over a refitted slope curve in unseen regions, ahead of tags in 16 of 17), while tags add $+0.4$ to $+0.9\%$ and neither vegetation, from 30 m maps or lidar point clouds, nor satellite embeddings match it (Sec. IV). We also give a Tobler-style function with one DEM-derived input ($+1.0\%$).

II. Data

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Sources. We use public GPS traces uploaded to OpenStreetMap (OSM) [17] in 17 mountain regions of the United States and Canada (Fig. 1), chosen for relief, trail density and lidar coverage. The traces come from three sources: the OSM trackpoints API, which returns the timestamped points of identifiable and trackable uploads tile by tile; the original GPX files of identifiable uploads, which give whole hikes at the device's own sampling; and a crawl of 1.5 million original files of public and identifiable uploads from 2016–2022, which keeps the timestamps the API drops. Sixteen regions combine all three; the Bay Area uses the crawl only, because the API times out on its tiles.

Processing. Each recording is cleaned (repeated timestamps, steps faster than 60 km/h), split at gaps over 5 min and thinned to points at least 5 s apart. A step counts as a stop when the point 60 s later lies within 10 m. Traces are map-matched to the OSM trail and road network with a hidden Markov model [19] (97% of points matched), and the matched path is cut into 50 m segments. Each segment records its moving speed, its signed slope along the direction of travel from the

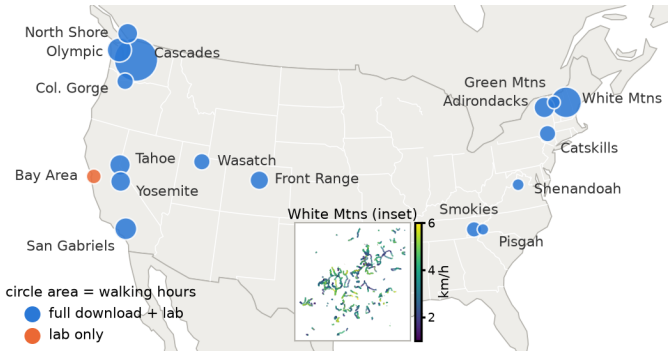


Fig. 1. The 17 regions. Circle area is walking hours (1,992 outings, 4,702 h in total). Blue: full OSM download plus the lab crawl; orange: lab crawl only. Inset: the White Mountains' 34,535 walking segments colored by speed.

5 m USGS 3DEP elevation grid [15] (resampled from 1 m lidar under at least 83% of walked trail in 13 regions), the OSM tags of its way and the trace's recording interval. Duplicate uploads are removed, and traces that start within 15 min and 1 km of each other form one outing (a group hike), the unit we resample. Travel mode is inferred per trace, anchored on trails closed to bikes and horses. Against the activity labels in 327 uploaders' files, 93% of labelled hikes are classed as walking, and 5% of labelled "walks" say bike or ski.

The dataset. The regions hold 2,876 hiking traces. The walking analysis set (walking traces on trails) has 290,640 segments (14,532 km) from 2,046 traces in 1,992 outings and 4,702 moving hours. Its uploaders are OSM mappers and enthusiasts rather than a random sample of hikers, and busy API tiles do not reach their oldest uploads. Traces logged more than 10 s apart read up to 11% slow, because straight lines between points cut corners and short stops hide between them; the gains we report are the same at every recording rate (Sec. IV).

III. Methods

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Target and covariates. We predict the distribution of $\ln v$, the log moving speed of a walking segment. Covariates come in six groups: (a) slope; (b) OSM tags: sac_scale, trail_visibility, surface, width, smoothness, mtb_scale, highway type and permitted modes; (c) terrain from elevation, read every 1 m along the segment's matched path: the spread (SD) and maximum of its 5 m pitches, the relief around an 11 m moving average and the side slope, from the 5 m grid and, where available, 1 m lidar, which adds STRIDE's ground roughness [11]; (d) 2021 land cover and tree canopy at 30 m [20]; (e) 2022 AlphaEarth satellite embeddings at the segment midpoint [16], [21]; (f) in the 15 regions with lidar point clouds, understory density (returns at 0.85–1.2 m among those below 1.2 m [12]), canopy cover and height within 3 m of the path. As a control, "hidden climbing" is the extra time Tobler's function predicts when the segment is walked as ten 5 m pieces instead of at its average grade.

TABLE I
CRPS skill (%) vs the refitted slope curve. Additive / trees; 95% intervals for unseen regions resample whole regions.

Model	New hikers	New 10 km	**Unseen region**
Tobler 1993	−12.8	−12.7	−12.5
Campbell 2022	−0.9	−0.9	−0.5
STRIDE 2024 (refit)	−17.2 (+0.2)	−17.3 (+0.2)	−16.8 (+0.2)
OSM tags	+1.3 / +1.5	+1.1 / +1.5	+0.9 / +0.4
30 m vegetation	+0.0 / +0.1	−0.0 / −0.1	−0.1 / −0.8
Embeddings	+1.4 / +2.6	+1.4 / +1.9	+1.1 / +1.1
Lidar vegetation†	+0.4 / +0.5	+0.4 / +0.4	+0.3 / +0.2
Terrain	+2.2 / +3.2	+2.1 / +2.8	**+2.1 / +2.8**
Terrain + emb. + tags	+3.7 / +5.1	+3.8 / +5.1	+3.0 / +3.9
Eq. (1), one input	+1.0	+1.0	+1.0

†The 15 regions with lidar point clouds.

Models. The baselines are Tobler's function [2], the published median curve of Campbell et al. [6], STRIDE [11] as published and with its form refitted, and our reference: a rounded, asymmetric slope curve refitted to the training data, $\ln v = a - k_{\pm}\left(\sqrt{(s+s_0)^2+e^2}-e\right)$, with k_+ above the peak and k_- below it. Each covariate group then enters two models: an additive one (ridge regression of the curve's residuals on one-hot bins: deciles of continuous covariates, one level per tag value plus "missing") and gradient-boosted trees on slope and the covariates. The terrain-adjusted hiking function keeps Tobler's form but lets peak speed and slope sensitivity depend on one DEM-derived number, the steppiness z:

$$\ln v = a_0 + a_1 z - (k_0 + k_1 z)\,|s + s_0|, \qquad (1)$$

with z the SD of the path's 5 m pitches divided by 0.10.

Predictive distributions and tests. Every model predicts its point estimate plus the quantiles of its residuals on held-out outings of the training data (training outings split 70/15/15 into fitting, early stopping and calibration), so calibration has to transfer too. Splits never separate a segment from its hike: 5 folds of outings (new hikers), 5 folds of 10 km UTM blocks (new places) and, the headline, leave one region out (17 folds); stricter checks hold out whole zones or half the continent. We report the CRPS [18] of $\ln v$ as skill against the refitted slope curve, $1-\mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{slope}}$, with 95% intervals that resample outings (whole regions for unseen regions); the coverage of 90% intervals; and the median error in moving time over whole hikes after their first kilometre. A reference shows how much is the hiker: the median residual of a hike's first kilometre, shrunk by a factor fitted on calibration outings (0.61), applied to the rest of the hike. Asymmetry is measured on out-and-back pairs, the same outing on the same OSM way in both directions (mean grade at least 3%), as the ratio of speed down to speed up.

IV. Results

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Tobler's function fails everywhere. Walking speed peaks near 4.2 km/h on level to gently descending trail, not at 6 km/h (Fig. 2a): Tobler's function is about 20% too fast within ±10% grade and 18% too slow on climbs steeper than 25%, and in

unseen regions it scores -12.5% (Table I), worse in all 17. Campbell et al.'s published median curve [6] is about 20% too fast on climbs steeper than 25%; it ties our refit on average (-0.5%) but ranges by region from -3.8% (White Mountains) to $+5.8\%$ (Shenandoah). Every model stays calibrated in unseen regions (90% intervals cover about 89% of segments).

Downhill is faster, but less than Tobler implies. For the same hiker on the same trail (2,763 out-and-back pairs from 1,145 outings; Fig. 2b), descending is 8% faster than climbing at 3–10% grade and 22–23% above 20%, not the 42% Tobler implies. Our pooled curve implies nearly the same ratios, so its asymmetry is not an artifact of who walks which slopes.

Map tags do not transfer. OSM tags improve on the slope curve for new hikers in known regions ($+1.3\%$ / $+1.5\%$, additive / trees) but lose a third to three quarters of that in unseen regions ($+0.9\%$ [0.1, 1.7] / $+0.4\%$ [-1.1, 1.6]).

Terrain transfers. Terrain from elevation gains nearly as much in unseen regions as within them, $+2.1\%$ [1.5, 3.1] / $+2.8\%$ [1.8, 4.1]. It beats tags in 16 of 17 held-out regions (Fig. 3b), and even a slope curve refitted on the held-out region's own other hikers ($+1.2\%$). Terrain minus tags is $+0.7$ to $+2.2$ points (95% interval), above zero in every bootstrap draw. About 60% of the gain is uneven grade inside the segment that one average slope hides (the hidden-climbing control alone gives $+1.2\%$ / $+1.6\%$). Terrain is neither a region proxy (within regions it adds $\sim$1.9 points to a learned offset per region) nor a hiker proxy (it adds $\sim$2.3 points to the hiker's own pace).

No other remote-sensing input transfers as well. 30 m land cover and canopy add nothing. AlphaEarth embeddings help within regions ($+1.4\%$ / $+2.6\%$) but less, and less reliably, in unseen ones ($+1.1\%$). Vegetation from the lidar point clouds helps within a region (White Mountains, new 10 km squares: $+3.6\% \rightarrow +5.0\%$ added to terrain) but adds only $+0.2$ points [-0.2, 0.5] to terrain in unseen regions. Published STRIDE [11] scores -16.8%; refitted, $+0.2\%$.

Robustness. The terrain gain holds for newer hikes (trained to 2019, tested on 2020 on: $+2.2\%$ / $+2.9\%$; tags $+0.9\%$), at every GPS recording rate (traces every $\leq$2 s, 2–10 s and $>$10 s: $+1.7\%$, $+2.4\%$ and $+1.9\%$ in unseen regions, additive) and with no neighbouring region in training: with a whole zone held out, $+1.9\%$ / $+2.7\%$ (tags $+0.7\%$ / -0.1%); from West to East and back, $+2.6\%$ / $+3.3\%$ (tags $+1.5\%$ / -0.3%).

Whole hikes: the hiker dominates. Over 1,354 hikes in held-out regions, the median error in moving time after the first kilometre is 13.6% for the slope curve and 13.0% with terrain, but 10.0% with the hiker's pace on that kilometre and 9.6% with both (CRPS skill $+8.4\%$; with terrain $+10.7\%$ / $+11.6\%$).

Other travel modes. The walking curve times a constant ($\times$2.5) predicts runners as well as their own curve, but not bikes and skis ($\sim 4\times$ walking speed downhill, $\sim 1.6\times$ uphill; own curve 3.8% better).

V. Discussion and Conclusion

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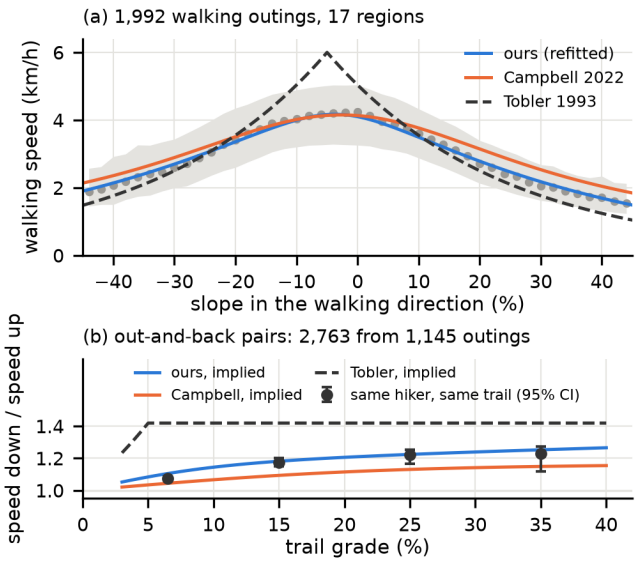


Fig. 2. (a) Walking speed vs slope in the walking direction: median per 2% bin (dots), middle 50% of segments (gray), our refitted curve, Campbell et al.'s median curve [6] and Tobler's function [2]. (b) Speed down / speed up for the same hiker on the same trail (out-and-back pairs; 95% intervals resample outings) and the ratio each curve implies.

Why tags do not transfer. Not because tag values mean different things in different places: within the same hike, harder sac_scale grades are slower in every region where they are used (mountain_hiking vs hiking -5%, 10 of 10 regions; harder grades -11%, 8 of 8). What varies is where tags exist: sac_scale is on 1–96% of walked trail, depending on the region, and untagged trail is faster than tagged trail in some regions and slower in others (-3.5% to $+5.1\%$ within a hike). So tags inform only part of each region's trails, and boosted trees on tags, which can fit region-specific tag combinations, lose the most in transfer ($+1.5\% \rightarrow +0.4\%$). Terrain exists on every trail, and its effect has the same sign in all 17 regions (steppiest vs smoothest fifth, -12%).

What terrain captures. Uneven grade within the segment, such as rock steps and short pitches, plus difficulty beyond grade. Its gain is largest on descents and level trail ($+2.3$ to $+2.8\%$, additive) and smallest on steep climbs ($+0.8\%$), consistent with footing setting the pace going down and effort going up. The signal is the trail's elevation profile, not micro-relief: 1 m lidar adds little over the 5 m grid, and STRIDE's off-path roughness and understory terms add nothing on trails or make it worse ($-18\% \rightarrow -35\%$).

Use. Eq. (1) needs only a DEM. Fitted on the other 16 regions, it gives a pace for all 3,369 km of White Mountains trail, 77% of which no trace in our data covers (median 15.2 min/km, vs 12.8 for Tobler). It travels less far than the full terrain model, though ($+0.6\%$ with a whole zone held out, -0.3% between East and West).

Limits. Our hikers are OSM uploaders, not a random sample. Sparse traces read slow (Sec. II); hiking functions depend on the sampling resolution [4]; distance along the

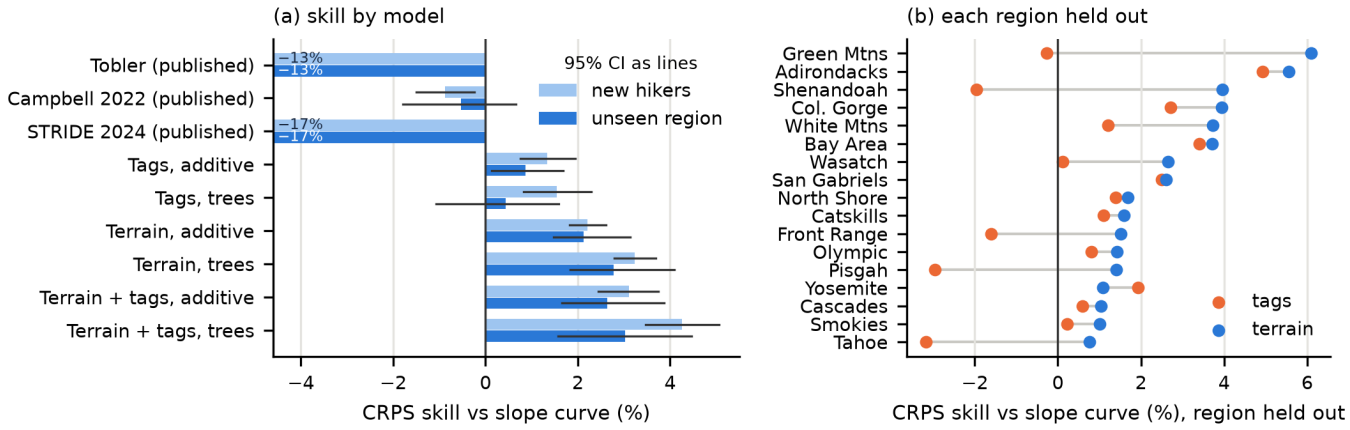


Fig. 3. CRPS skill vs the refitted slope curve. (a) By model, for new hikers in known regions (light) and in regions the model never saw (dark), with 95% intervals resampling whole regions. (b) Each region held out: terrain vs OSM tags (additive models); terrain beats tags in 16 of 17.

mapped trail would fix the cut corners and raise speeds by about 1–3%, and the comparisons hold at every recording rate. Per-segment gains are a few percent because GPS noise and the hiker dominate. Weather [22] (a factor in [8]) slows our hikers on hot days (-3% above 25°C) and with snow cover (-4%), but neither weather (-0.0%) nor date and time of day improves predictions in new regions, and once a hiker's pace on the day is known, weather adds nothing.

Conclusion. Across 17 North American regions, terrain from elevation data predicts hiking speed in regions a model has never seen; OSM tags, vegetation and satellite embeddings mostly help where it was trained. Tobler's function is too fast on gentle trail and overstates the speed-up downhill. A Tobler-style function with one DEM-derived input keeps part of the terrain gain and needs only a DEM, though a hiker's own pace matters most. [TODO: Data release (segments without user names) if the mentors agree.]

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